

# Improving the Unconditional Constant in Loo–Perreau’s Non-SCD Renorming

Automated Math Discovery Smoke Test

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## Source and Open Question

This packet concerns Question 5.4 from:

Marcus Loo and Yoel Perreau, *A Banach space with an unconditional basis which is not slicely countably determined*, arXiv:2603.12947.

The question appears in Section 5, page 18 of the source paper:

$$\overline{\bigcup_{k \in A} B_k} \quad \overline{\bigcup_{k \in A} B_k}$$

Summing up, we have shown that

$$\|x + \theta y_\alpha\| \leq \max\left\{\frac{3}{2} + 2\varepsilon, \frac{1}{2}(|1 + \theta| + |1 - \theta|) + 4\varepsilon\right\} \leq \max\left\{\frac{3}{2} + 2\varepsilon, \sqrt{2} + 4\varepsilon\right\}.$$

So if  $\varepsilon$  was initially chosen small enough, we would get that this quantity is strictly smaller than 2, and therefore  $x$  is not a super alternative Daugavet point, as claimed.  $\square$

In Subsection 3.3, we produced an example of a Banach space with a 2-unconditional basis  $Y$  whose unit ball fails to be SCD. To the best of our knowledge, the following is unknown.

**Question 5.4.** Can we produce a Banach space with a  $k$ -unconditional basis for  $k \in (1, 2)$  whose unit ball is not SCD?

Also, observe that the unit ball of the latter space  $Y$  does contain many SCD points (since  $\overline{\text{conv}}(\Pi^+ \cup \Pi^-) \subseteq \text{SCD}(B_Y)$ ). So it is natural to ask whether it is possible to do a little bit better.

**Question 5.5.** Does there exist a Banach space with an unconditional basis whose unit ball has no SCD point (equivalently, such that 0 is not an SCD point in this set)?

The alert reader might wonder what would happen if we were simply to remove the strongly exposed points from the above set, or in other words to consider the symmetrization  $D$  of the set  $\Omega^+ = \Sigma^+ \setminus \Pi^+$  instead of  $C$ . It turns out that 0 is still SCD in this new unit ball.

The source paper constructs, from the binary tree space  $X = X_T$ , an equivalent norm with a 2-unconditional basis whose unit ball is not SCD. Question 5.4 asks whether the constant can be improved to any  $k \in (1, 2)$ .

## Imported Notation and Facts

We use the notation of the source paper. Thus  $X = X_T$  is the binary tree space,  $(e_t)_{t \in T}$  is its canonical 1-unconditional basis,  $B_X$  is its original unit ball, and

$$B_X^+ = \{x \in B_X : x(t) \geq 0 \text{ for all } t \in T\}, \quad B_X^- = -B_X^+.$$

The paper defines  $\Sigma$  and  $\Pi$  inside  $B_X$  and writes  $\Omega = \Sigma \setminus \Pi$ . We need two facts proved there:

1.  $\Omega^+$  is non-empty and contained in  $S_X^+$ .
2. No point  $x \in \Omega^+$  is an SCD point of  $B_X^+$ . This is Proposition 3.9 of the source paper.

We also use the standard localized form of the SCD-point criterion: for a bounded convex set, countable determining families of non-empty relatively weakly open subsets are equivalent to countable determining families of slices. This is the pointwise version of the Bourgain-lemma reduction used in the source paper for SCD sets.

## Main Claim

**Theorem 1.** *For every  $k \in (1, 2)$  there is a Banach space with a  $k$ -unconditional basis whose unit ball is not slicely countably determined.*

*Proof.* Fix  $t \in (1/2, 1)$  and set

$$C_t = \overline{\text{conv}}(B_X^+ \cup B_X^- \cup tB_X).$$

The set  $C_t$  is closed, convex, symmetric, and absorbing; hence it is the unit ball of an equivalent norm, which we denote by  $\|\cdot\|_t$ . Moreover

$$tB_X \subseteq C_t \subseteq B_X.$$

Consequently

$$\|x\| \leq \|x\|_t \leq t^{-1} \|x\| \quad (x \in X).$$

We first estimate the unconditional constant of the canonical basis in  $(X, \|\cdot\|_t)$ . Let  $D_\varepsilon$  be any sign-change operator,  $D_\varepsilon(\sum a_t e_t) = \sum \varepsilon_t a_t e_t$ , where  $\varepsilon_t \in \{-1, 1\}$ . Since the original basis of  $X$  is 1-unconditional and  $C_t \subseteq B_X$ , we have

$$D_\varepsilon(C_t) \subseteq D_\varepsilon(B_X) = B_X.$$

But  $tB_X \subseteq C_t$ , so  $B_X \subseteq t^{-1}C_t$ . Hence

$$D_\varepsilon(C_t) \subseteq t^{-1}C_t.$$

Thus  $\|D_\varepsilon\|_{(X, \|\cdot\|_t) \rightarrow (X, \|\cdot\|_t)} \leq t^{-1}$  for every choice of signs, so the basis is  $t^{-1}$ -unconditional.

It remains to show that  $C_t$  is not SCD. Choose  $x \in \Omega^+$ . We will show that  $x$  is not an SCD point of  $C_t$ .

Assume toward a contradiction that  $(S_n)$  is a countable sequence of slices of  $C_t$  determining  $x$ . Let

$$M = \{n \in \mathbb{N} : S_n \cap B_X^+ \neq \emptyset\}.$$

For  $n \in M$ , put  $U_n = S_n \cap B_X^+$ . We claim that the family  $(U_n)_{n \in M}$  determines  $x$  inside  $B_X^+$ .

If  $M = \emptyset$ , then, arguing as below, choose  $z_n \in S_n \cap (B_X^- \cup tB_X)$  for every  $n$ . The determining property would give  $x \in \overline{\text{conv}}\{z_n : n \in \mathbb{N}\}$ . Applying a branch functional  $\beta$  with  $\beta(x) = 1$  is impossible, since  $\beta(z_n) \leq t < 1$  for all  $n$ . Thus  $M$  is non-empty. If  $\mathbb{N} \setminus M = \emptyset$ , the claim is immediate from the fact that the original slices determine  $x$  in  $C_t$ . So assume from now on that both  $M$  and  $\mathbb{N} \setminus M$  are non-empty.

Take an arbitrary selection  $y_n \in U_n$  for  $n \in M$ . If  $n \notin M$ , then  $S_n$  does not meet  $B_X^+$ . Since  $C_t = \overline{\text{conv}}(B_X^+ \cup B_X^- \cup tB_X)$ , each such slice meets  $B_X^- \cup tB_X$ ; choose  $z_n$  there. Because the slices  $(S_n)$  determine  $x$  in  $C_t$ , the point  $x$  belongs to the closed convex hull of the selected points  $y_n$  and  $z_n$ .

Let  $\varepsilon > 0$ . We may therefore find

$$y \in \text{conv}\{y_n : n \in M\}, \quad z \in \text{conv}\{z_n : n \notin M\}, \quad \lambda \in [0, 1],$$

such that, in the original norm of  $X$ ,

$$\|x - (\lambda y + (1 - \lambda)z)\| < \varepsilon.$$

Pick a branch  $\beta$  of the binary tree for which

$$\sum_{s \in \beta} x(s) = 1.$$

The corresponding branch functional has norm one on  $X$ . Since  $y \in B_X^+$ , we have  $\sum_{s \in \beta} y(s) \leq 1$ . Since  $z \in \text{conv}(B_X^- \cup tB_X)$ , we have  $\sum_{s \in \beta} z(s) \leq t$ . Applying the branch functional gives

$$(1 - \lambda)(1 - t) \leq 1 - \lambda \sum_{s \in \beta} y(s) - (1 - \lambda) \sum_{s \in \beta} z(s) \leq \|x - (\lambda y + (1 - \lambda)z)\| < \varepsilon.$$

Therefore  $1 - \lambda < \varepsilon/(1 - t)$ . Since  $y, z \in B_X$ ,

$$\|x - y\| \leq \|x - (\lambda y + (1 - \lambda)z)\| + (1 - \lambda) \|y - z\| < \varepsilon + \frac{2\varepsilon}{1 - t}.$$

As  $\varepsilon$  was arbitrary,  $x \in \overline{\text{conv}}\{y_n : n \in M\}$ . Thus the relatively weakly open family  $(U_n)_{n \in M}$  determines  $x$  in  $B_X^+$ .

By the localized SCD-point criterion,  $x$  would be an SCD point of  $B_X^+$ , contradicting Proposition 3.9 of Loo-Perreau. Hence  $C_t$  is not SCD.

Finally, given  $k \in (1, 2)$ , take  $t = 1/k$ . Then  $t \in (1/2, 1)$ , the canonical basis of  $(X, \|\cdot\|_t)$  is  $k$ -unconditional, and its unit ball  $C_t$  is not SCD.  $\square$

## Verification Notes

The only numerical parameter check is

$$k \in (1, 2) \iff t = 1/k \in (1/2, 1),$$

and the branch-functional estimate loses the factor  $(1 - t)^{-1}$ , which is finite precisely because  $t < 1$ . The short script `code/check_interpolation_constants.py` verifies these inequalities for sample values and records the constant calculation used above.

**Remark 1.** *The result does not address Question 5.5 of the source paper. The construction still uses the same family  $\Omega^+$  of non-SCD points, but it does not show that the whole unit ball has no SCD points.*